Exercice 21

$t^2x'' + atx' + bx = g(t)$

a)

Pour $t > 0$

Posons $t = e^u$, $u = \ln(t)$

$(e^u)^2(x(e^u))'' + ae^u(x(e^u))' + bx(e^u) = g(e^u)$

$x(t)' = \frac{dx}{dt} = \frac{dx}{du} \cdot \frac{du}{dt}$

$x' = \frac{1}{t}\frac{dx}{du} = \frac{1}{e^u} \cdot \frac{dx}{du}$

$x'' = \frac{d^2x}{dt^2} = \frac{d(x')}{dt} = \frac{d(x')}{du} \cdot \frac{du}{dt}$

$= \frac{d\left(\frac{1}{e^u} \cdot \frac{dx}{du}\right)}{du} \cdot \frac{1}{e^u}$

$x(e^u)'' = \frac{1}{e^u}\Big(\frac{d(}{d}$

$= \frac{1}{e^{2u}}\Big($

$= \frac{1}{e^{2u}}$

$e^{2u}\Big(\frac{1}{e^{2u}}\Big(\frac{d^2}{du}$

$ae^u\Big(\frac{1}{e^u}$

$\frac{d^2x}{du^2} - \frac{dx}{du}$

$)$

$g(e^u)$

$\frac{1}{e^u}$

$$x(e^u)'' = \frac{1}{e^u}\left(\frac{d(\frac{1}{e^u})}{du}\cdot\frac{dx}{du} + \frac{1}{e^u}\frac{d}{du}\left(\frac{dx}{du}\right)\right)$$

$$= \frac{1}{e^u}\left(-\frac{1}{e^u}\frac{dx}{du} + \frac{1}{e^u}\frac{d^2x}{du^2}\right)$$

$$= \frac{1}{e^{2u}}\left(\frac{d^2x}{du^2} - \frac{dx}{du}\right)$$

$$e^{2u}\left(\frac{1}{e^{2u}}\left(\frac{d^2x}{du^2} - \frac{dx}{du}\right)\right) +$$

$$ae^u\left(\frac{1}{e^u}\frac{dx}{du}\right) + bx(e^u) = g(e^u)$$

$$\frac{d^2x}{du^2} - \frac{dx}{du} + a\frac{dx}{du} + bx(e^u) = g(e^u)$$

$\frac{d^2x}{du^2} + ($

b-) Rés

Pour a

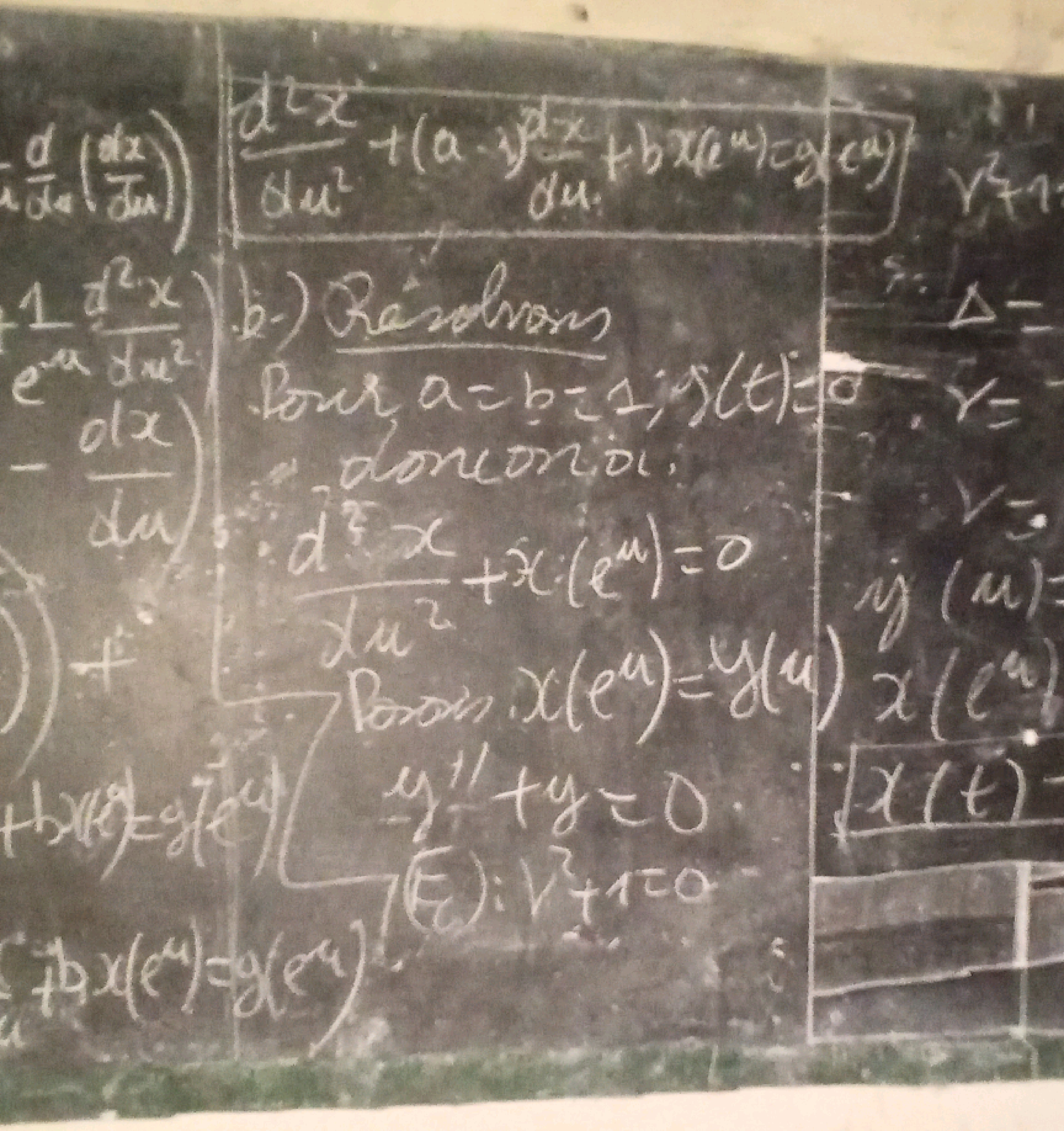
alor

$\frac{d^2x}{du^2}$

Posoi

y'

(E_c)



$\frac{d}{du}\left(\frac{dx}{du}\right)$

$\frac{1}{e^u}\frac{d^2x}{du^2}$

$-\frac{dx}{du}$

$+b\,x(e^u)=g(e^u)$

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$$\boxed{\frac{d^2x}{du^2}+(a-1)\frac{dx}{du}+b\,x(e^u)=g(e^u)}$$

b-) Résolvons

Pour $a=b=1$; $g(t)=0$

donc on a :

$$\frac{d^2x}{du^2}+x(e^u)=0$$

Posons $x(e^u)=y(u)$

$$y''+y=0$$

$(E_c) : r^2+1=0$

r^2+1

$\Delta =$

$r =$

$r =$

$y(u) =$

$x(e^u)$

$\boxed{x(t)}$

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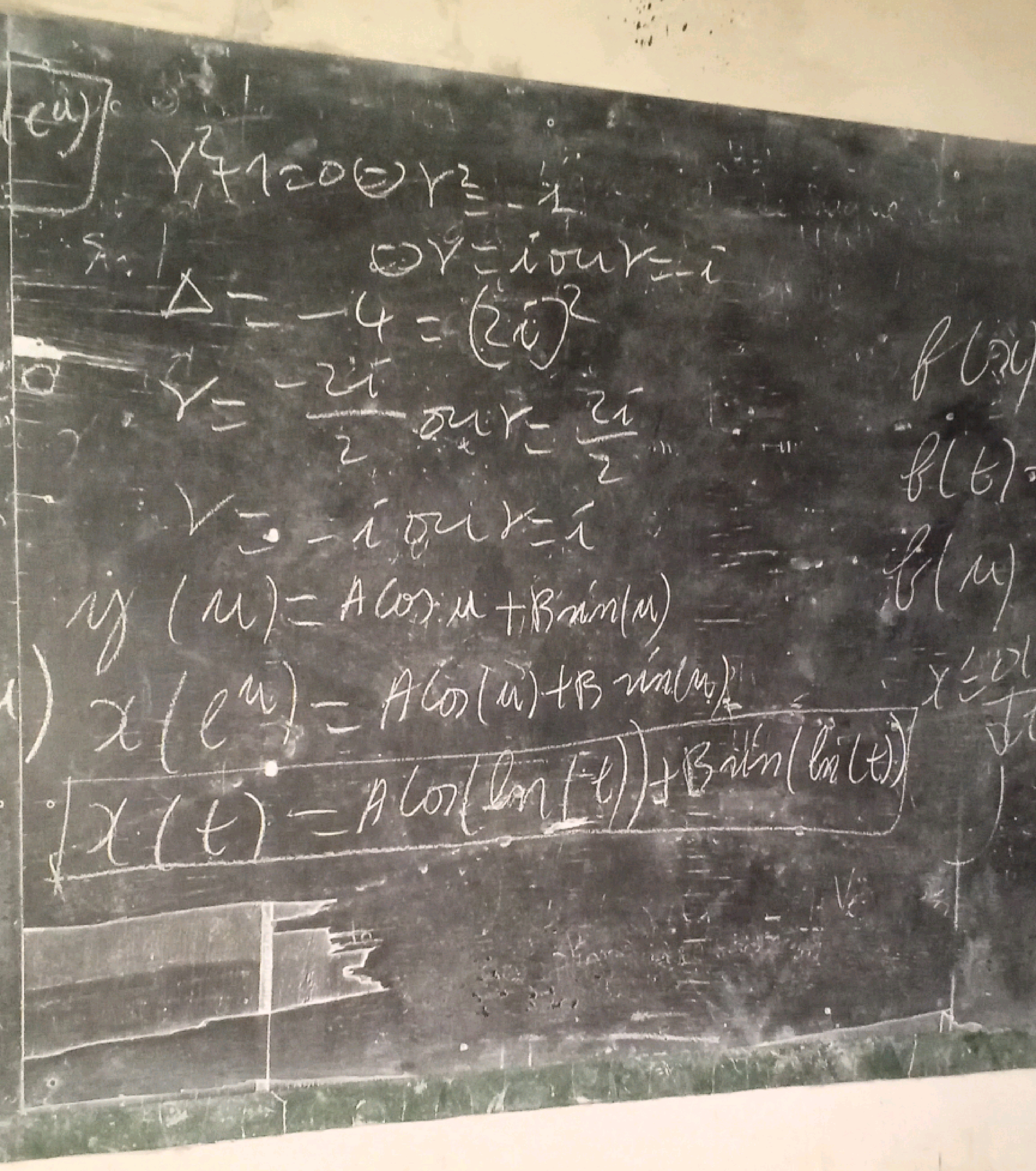

$\boxed{(e^{u})}$

$r^2+1=0 \Leftrightarrow r^2=-1$

$\Leftrightarrow r=i$ ou $r=-i$

$\Delta=-4=(2i)^2$

$r=\frac{-2i}{2}$ ou $r=\frac{2i}{2}$

$r=-i$ ou $r=i$

$y(u)=A\cos u+B\sin(u)$

$x(e^{u})=A\cos(u)+B\sin(u)$

$\boxed{x(t)=A\cos(\ln(t))+B\sin(\ln(t))}$

$f(x)$

$f(t)=$

$f(u)$

$x=$

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